

2. Derive the gradient of MSE Cost Function $J(m, b)$.

$$J(m, b) = \frac{1}{n} \sum (\hat{y} - y)^2 \quad \text{But } n=2$$

$$= \frac{1}{2} \sum (\hat{y} - y)^2$$

$$= \frac{1}{2} \cdot (\hat{y} - y)^2$$

$$\text{but } e = \hat{y} - y$$

Therefore;

$$J = \frac{1}{2}$$

Therefore, replace $\hat{y} - y$ with e .

$$J = \frac{1}{2} \sum e^2$$

But we have 2 number of samples. Therefore;

$$J = \left(\frac{1}{2}\right) \times (e_1^2 + e_2^2)$$

$$\text{But } e_1 = m_1 x_{11} + m_2 x_{21} + b_1 - y_1$$

$$e_2 = m_1 x_{12} + m_2 x_{22} + b_2 - y_2$$

$$\begin{aligned} \text{Using Our data, } e_1 &= m_1 \cdot 1 + m_2 \cdot 3 + b_1 - 5 \\ &= m_1 + 3m_2 + b_1 - 5 \\ &= 3m_1 + m_2 + b_1 - 5 \end{aligned}$$

$$\begin{aligned} e_2 &= m_1 \cdot 4 + m_2 \cdot 10 + b_2 - 6 \\ &= 4m_1 + 10m_2 + b_2 - 6 \\ &= 40m_1 + m_2 + b_2 - 6 \end{aligned}$$

$$\text{Compute } dJ/dm_1 = \frac{dJ}{de_1} \times \frac{de_1}{dm_1} + \frac{dJ}{de_2} \times \frac{de_2}{dm_1}$$

$$\text{Computed } \frac{dJ}{de_1} : J = \frac{1}{2} (e_1^2 + e_2^2)$$

$$\frac{dJ}{de_1} = \frac{1}{2} \times 2e_1$$

$$\frac{dJ}{de_1} = e_1$$

Compute $\frac{de_1}{dm_1}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$\frac{de_1}{dm_1} = 3m_2 \rightarrow$ Only m_1 varies, everything else is constant

Compute $\frac{dJ}{de_2}$: $\frac{dJ}{de_2} = \frac{1}{2} \cdot 2e_2$
 $\frac{dJ}{de_2} = e_2$

Compute $\frac{de_2}{dm_1}$: $e_2 = 40m_1 m_2 + b_2 - 6$
 $\frac{de_2}{dm_1} = 40m_2$

Combine using chain rule: $\frac{dJ}{dm_1} = e_1 \times 3m_2 + e_2 \times 40m_2$
 $= m_2 (3e_1 + 40e_2)$

$\therefore \frac{dJ}{dm_1} = m_2 (3e_1 + 40e_2)$

For dJ/db_2 :

Compute $\frac{de_2}{db_2}$: $e_2 = 40m_1 m_2 + b_2 - 6$
 $\frac{de_2}{db_2} = 1$

Combine using chain rule

$\frac{dJ}{db_2} = \frac{dJ}{de_2} \times \frac{de_2}{db_2}$
 $= e_2 \times 1$

$\frac{dJ}{db_2} = e_2$

For dJ/dm_2

Compute $\frac{de_1}{dm_2}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$$\frac{de_1}{dm_2} = 3m_1$$

Compute $\frac{de_2}{dm_2}$: $e_2 = 40m_1 m_2 + b_2 - 6$

$$\frac{de_2}{dm_2} = 40m_1$$

Combine using chain rule

$$\frac{dJ}{dm_2} = \frac{dJ}{de_1} \times \frac{de_1}{dm_2} + \frac{dJ}{de_2} \times \frac{de_2}{dm_2}$$

$$= e_1 \times 3m_1 + e_2 \times 40m_1$$

$$= m_1 (3e_1 + 40e_2)$$

$$\therefore \frac{dJ}{dm_2} = m_1 (3e_1 + 40e_2)$$

For dJ/db_1

Compute $\frac{de_1}{db_1}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$$\frac{de_1}{db_1} = 1$$

Combine using chain rule:

$$\frac{dJ}{db_1} = \frac{dJ}{de_1} \times \frac{de_1}{db_1}$$

$$= e_1 \times 1$$

$$\therefore \frac{dJ}{db_1} = e_1$$